

To disentangle these two explanations, we introduce a notion of *myopic* transformer models, which we show to be incapable of deliberate pre-caching—for these models, the extent to which past features are beneficial to the future is decided purely by the breadcrumbs explanation. Thus, the gap between vanilla and myopic transformer models is a quantitative measure of how much pre-caching is taking place.

3.1 Causal sequence modeling

Suppose, for the sake of exposition, that the transformer model G uses independent parameters for each position.³ Let p be the parameter count of each forward pass of G . Then, letting $\theta_i \in \Theta = \mathbb{R}^p$ be all parameters used by G at position i , a transformer G is a parameterized function

$$G: \mathbb{X}^n \times \Theta^n \rightarrow \mathbb{Y}^n, \quad (\mathbf{x}_1, \dots, \mathbf{x}_n; \theta_1, \dots, \theta_n) \mapsto (\hat{\mathbf{y}}_1, \dots, \hat{\mathbf{y}}_n).$$

For $1 \leq i \leq n$, let $G_i(\mathbf{x}_1, \dots, \mathbf{x}_n) \in \mathbb{Y}$ be the output of G 's i th forward pass. Because of the causal masking within G , this depends only on $\mathbf{x}_1, \dots, \mathbf{x}_i$ and $\theta_1, \dots, \theta_i$. That is, with slight abuse of notation, we may write

$$\hat{\mathbf{y}}_i = G_i(\mathbf{x}_1, \dots, \mathbf{x}_n; \theta_1, \dots, \theta_n) = G_i(\mathbf{x}_1, \dots, \mathbf{x}_i; \theta_1, \dots, \theta_i).$$

3.2 Off-diagonal gradient terms

Now, letting $\mathcal{L}: \mathbb{Y} \times \mathbb{Y} \rightarrow \mathbb{R}_+$ be some choice of loss function, the expected loss ℓ of a transformer model with parameters $\theta_1, \dots, \theta_n$ is

$$\ell(\theta_1, \dots, \theta_n) := \mathbb{E}_{(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \sim \mathcal{D}} \sum_{i=1}^n \mathcal{L}(G_i(\mathbf{x}_1, \dots, \mathbf{x}_i; \theta_1, \dots, \theta_i), \mathbf{y}_i) =: \sum_{i=1}^n \ell_i(\theta_1, \dots, \theta_i),$$

the sum over $1 \leq i \leq n$ of the expected loss ℓ_i at position i . (We suppress the dependence on G and $\mathcal{D}$ for concision.) In practice, we always tie the weights across position. That is, all θ_i are set equal to the same $\theta \in \Theta$. Then, by the chain rule,

$$\nabla_{\theta} \ell(\theta, \dots, \theta) = \sum_{i=1}^n \nabla_{\theta_i} \ell(\theta_1, \dots, \theta_n) \big|_{\theta_1=\dots=\theta_n=\theta} = \sum_{i=1}^n \sum_{j=i}^n \nabla_{\theta_i} \ell_j(\theta_1, \dots, \theta_j) \big|_{\theta_1=\dots=\theta_n=\theta},$$

a sum over an upper-triangular expected Jacobian “matrix”. The off-diagonal terms $i < j$, corresponding to the expected gradient of the model’s future loss at position j with respect to its weights at position i , are the training signals that encourage pre-caching.

3.3 Measuring pre-caching: The myopia gap

We say a model is *myopic* when each forward pass G_i optimizes only ℓ_i without regard for future ℓ_j at $j > i$. In the untied weights case, the right definition is then apparent.

Definition 1. The parameters $(\tilde{\theta}_1, \dots, \tilde{\theta}_n) \in \Theta^n$ are *untied-myopic* if they satisfy

$$\tilde{\theta}_i \in \arg \min_{\theta_i} \ell_i(\tilde{\theta}_1, \dots, \tilde{\theta}_{i-1}, \theta_i) \quad \forall i \in \{1, \dots, n\}. \quad (1)$$

Definition 2. Let $\mathbb{M}$ be the feasible set of the constraints in Equation 1. The *untied myopia gap* is the smallest possible gap between the expected loss attained by a myopic model and the optimal model:

$$p^* := \min_{(\tilde{\theta}_1, \dots, \tilde{\theta}_n) \in \mathbb{M}} \ell(\tilde{\theta}_1, \dots, \tilde{\theta}_n) - \min_{(\theta_1, \dots, \theta_n) \in \Theta^n} \ell(\theta_1, \dots, \theta_n) \geq 0. \quad (2)$$

In the tied weights case, it is perhaps not immediately clear what the right definition of myopia should be. It does not suffice to simply constrain the minimizations in Equation 1 to $\tilde{\theta}_1 = \dots = \tilde{\theta}_n$, since $\min_{\theta} \ell_i(\theta, \dots, \theta)$ is optimizing for pre-caching (the dependence on arguments $j < i$) as well as the present inference (the dependence on argument i). Instead,

³For example, this is true of absolute position embedding weights.

the right notion is a choice of tied parameters such that the model is, aggregated over positions, optimal for the present task when conditioned on a fixed past. That is, forward passes do not compute features for the future if they can compute other features more beneficial to the present.

Definition 3. The parameters $\tilde{\theta} \in \Theta$ are **(tied-)myopic** if they satisfy

$$\tilde{\theta} \in \arg \min_{\theta \in \Theta} \sum_{i=1}^n \ell_i(\tilde{\theta}, \dots, \tilde{\theta}, \theta). \quad (3)$$

The **(tied) myopia gap** is then defined analogously to Definition 2.

The *breadcrumbs hypothesis* states that the myopia gap is small—near-optimal performance can be attained even when each forward pass is computing features relevant to only its own immediate inference task, with no regard to pre-caching for the future.

If the breadcrumbs hypothesis does not hold, we say that the model is *pre-caching*. It is important to remember that the ℓ_i depend on a choice of transformer model G and dataset $\mathcal{D}$. That is, breadcrumbs and pre-caching are properties of the model architecture and the data considered as a whole.

Although a small myopia gap reveals that one can do just as well without pre-caching, it does not say much about any specific model. To measure pre-caching within a given model, we examine the extent to which its parameters violate the myopia constraints.

Definition 4. The **(tied) local myopia bonus** at $\theta^* \in \Theta$ is

$$\hat{c}(\theta^*) := \max_{\theta \in \Theta} \sum_{i=1}^n (\ell_i(\theta^*, \dots, \theta^*) - \ell_i(\theta^*, \dots, \theta^*, \theta)) \geq 0.$$

For further interpretation of the myopia gap and myopia bonus, see Appendix A.

3.4 Myopic gradient descent

Our heuristic remark in Section 3.2, that the off-diagonal gradient terms are responsible for pre-caching, is justified by Theorem 13 below. It states that, given certain regularity conditions on the loss terms ℓ_i , performing gradient descent with the off-diagonal terms removed results in a myopic model in the sense of Definition 3. We call this *myopic descent*.

For myopic descent to be stable in the tied-weights case, we need, roughly speaking, for the model to depend more on the parameters associated with the present forward pass than those from the past. This is a plausible condition—dependence on the past is mediated purely by the attention mechanism, while the present forward pass depends on both attention and feedforward parameters. The precise condition we use is **forward bias** (Definition 11); see Appendix C for details and the proof of the theorem.

Theorem 13. Let $f(\tilde{\theta}, \theta) := \sum_{i=1}^n \ell_i(\tilde{\theta}, \dots, \tilde{\theta}, \theta)$. If f is forward-biased, σ -strongly convex, and L -smooth, then, for some step size $\eta > 0$, the iterates of myopic descent with tied weights

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_{\theta} f(\tilde{\theta}, \theta) \big|_{\tilde{\theta}=\theta=\theta^{(t)}}$$

converge to $\tilde{\theta} \in \Theta$ satisfying the myopia constraints of Equation 3.

4 Synthetic data experiments

4.1 The $\mathcal{D}_p$ task

To demonstrate a simple example where the transformer model learns to pre-cache (and thus the myopia gap is large), we construct the following synthetic dataset.

Definition 5. The data distribution $\mathcal{D}_{p,a,b}^N$ is defined as the joint distribution of real-valued random variables $(x_n)_{n=1}^N, (y_n)_{n=1}^N, (z_n)_{n=1}^N$ where, for each n ,

- $x_n \sim \mathcal{N}(0, 1)$ (standard Gaussian)
- $z_n \sim \text{Ber}(p)$ (Bernoulli with probability p)
- $y_n = z_n \sum_{i=1}^a \sin(bx_{n-i}) + (1 - z_n)x_n$

and $\{x_n\}_{n \in \mathbb{N}} \cup \{z_n\}_{n \in \mathbb{N}}$ are mutually independent. In our experiments, we always set the parameters $a = b = 10$ and $N = 64$, so for convenience notate $\mathcal{D}_p := \mathcal{D}_{p,10,10}^{64}$.

The intuition is that a transformer regression model G trained on $\mathcal{D}_p$ would benefit from pre-caching $\sin(bx_n)$ during its forward pass at position n , even though this computation is irrelevant to its task of predicting y_n . One simple strategy that makes use of this pre-caching is Algorithm 1.⁴

The motivation for the Bernoulli variables z_i is that, as p decreases, the expected first time when $\sin(bx_n)$ becomes useful advances further into the future. In addition, when p is sufficiently small, the probability $(1 - p)^a$ that the value $\sin(bx_n)$ is never useful at all becomes non-negligible. We will show that, even in this case, the transformer model learns to pre-cache.

Investigating myopia. Suppose that we train a myopic model (Section 3.4) on the same task. Since this model lacks off-diagonal gradient terms, we do not expect it to learn to pre-cache $\sin(bx_n)$ at position n . One possible strategy that does not use pre-caching is Algorithm 2. We expect this brute force algorithm to perform significantly worse given the same parameter count—it computes an a -dimensional nonlinear function within a single layer, while each layer of Algorithm 1 computes only scalar nonlinear functions.⁵

Algorithm 1 Pre-caching algorithm

At position n ,
input x_n, z_n
layer 1 compute $F_n := \sin(bx_n)$
layer 2 read F_{n-i} for $i = 1, \dots, a$.
layer 2 compute
 $\hat{y}_n := z_n \sum_{i=1}^a F_{n-i} + (1 - z_n)x_n$.
return $\hat{y}_n$.

Algorithm 2 Brute force algorithm

At position n ,
input x_n, z_n
layer 1 compute $\emptyset$
layer 2 read x_{n-i} for $i = 1, \dots, a$.
layer 2 compute
 $\hat{y}_n := z_n \sum_{i=1}^a \sin(bx_{n-i}) + (1 - z_n)x_n$.
return $\hat{y}_n$.

4.1.1 Evaluation: linear probing

To determine if the transformer model is computing $\sin(bx_n)$ at position n , we fit linear probes on the hidden states. We additionally compute the correlations between $\sin(bx_n)$ and each individual dimension (i.e., each neuron) of each hidden state.⁶ See Section D.1.1 for details.

⁴We think of each transformer layer as a **read** operation, performed by the attention mechanism, followed by a **compute** operation, performed by the feedforward block.

⁵For example, Shen et al. (2022) provide upper bounds on MLP error that degrade exponentially in dimensionality given a fixed parameter and layer count.

⁶Note that, in order for linear probing to be meaningful, we must first ensure that there is no pre-existing linear relationship between the inputs and the quantities we are probing for. Since the $(x_n, z_n)_n$ are mutually independent, this follows from Lemma 15, stating that x_n and $\sin(bx_n)$ have near-zero correlation for large enough b . In our experiments, we set $b = 10$, in which case $\rho(x_n, \sin(bx_n)) < 10^{-20}$. In other words, there is low predictive $\mathcal{V}$ -information from the inputs to the target $\sin(bx_n)$, where $\mathcal{V}$ is the class of linear models (Xu et al., 2020).